

IT24100327 Perera B.P.N

PS – Lab 07

1. A train arrives at a station uniformly between 8:00 a.m. and 8:40 a.m. Let the random variable X represent the number of minutes the train arrives after 8:00 a.m. What is the probability that the train arrives between 8:10 a.m. and 8:25 a.m.?

```
> punif(25,min = 0, max = 40, lower.tail = TRUE) - punif(10, min = 0, max = 40$  
[1] 0.375
```

2. The time (in hours) to complete a software update is exponentially distributed with rate $\lambda = 1/3$. Find the probability that an update will take at most 2 hours.

```
> pexp(2,rate = 1 / 3, lower.tail = TRUE)  
[1] 0.4865829
```

3. Suppose IQ scores are normally distributed with a mean of 100 and a standard deviation of 15.
 - I. What is the probability that a randomly selected person has an IQ above 130?

```
> pnorm(130, mean = 100, sd = 15, lower.tail = FALSE)  
[1] 0.02275013
```

- II. What IQ score represents the 95th percentile?
- III.

```
> qnorm(0.95, mean = 100, sd = 15, lower.tail = TRUE)  
[1] 124.6728
```